

HONGYI XIE

Research Fellow, Nanyang Technological University, Singapore

Email: hyxie22@m.fudan.edu.cn; Website: <https://xieeco.github.io>

Google Scholar: <https://scholar.google.com/citations?user=7LgFC9wAAAAJ>

EMPLOYMENT

Research Fellow, Asian School of the Environment, Nanyang Technological University Jul 2026 -now
Engineer, Transport Planning and Research Institute, Ministry of Transport Jul 2020 - Jul 2022
Research assistant, National Taiwan University Jul 2015 - Sep 2015

EDUCATION

Fudan University Sep 2022 - Jun 2026
Ph.D., with honors, Environmental Management
National University of Singapore Aug 2024 - Aug 2025
Joint Ph.D., Civil and Environmental Engineering
Beijing Normal University Sep 2017 - Jun 2020
M.S., with honors, Environmental Science
Zhejiang University Sep 2013 - Jun 2017
B.S., Environmental Science

RESEARCH INTEREST

Co-benefits of cross-sector climate policies; Constraints on renewable energy scale-up; Energy-climate-environment nexus

PUBLICATIONS

denotes equal contribution; * denotes the corresponding author.

Peer reviewed:

Selected:

1. Xie, H.[#], Wu, Y.[#], Wang, Y.^{*}, Guo, Y.^{*}. (2026). Addressing water resource constraints for electrolytic hydrogen demand in China. *Nature Sustainability*. <https://doi.org/10.1038/s41893-026-01894-9>
2. Xie, H., Chen, B.^{*}, Wang, C., Xie, W., Long, X., Zhang, D., Dai, M., Han, Z., Song, Y., Li, J., Wang, Y.^{*}. (2025). Uncertain cross-sector climate actions could undermine air pollution reduction co-benefits in China's power and passenger car sectors. *Engineering*, In press. <https://doi.org/10.1016/j.eng.2025.09.006>

3. **Xie, H.**, Chen, B., Dai, M., Han, Z., Bai, Y., Wang, Y.^{*}. (2024). Upgrading passenger vehicle emission standard help to reduce China's air pollution risk from uncertainty in electrification. *Environmental Science & Technology*, 58, 5325-5335. (Supplementary cover paper). <https://doi.org/10.1021/acs.est.3c10078>
4. **Xie, H.**, Li, H., Wang, S., Han, Z., Chen, B., Lv, H., Wang, Y.^{*}. Shifting retired battery disposal from metallurgical recycling to high-value utilization for optimizing decarbonization in China. *Environmental Science and Ecotechnology*. In press.

Others:

19. Wang, C., Xue, J., **Xie, H.**, Phillips, F., Liu, G., Zhang, F.^{*}. (2026). Tiered standards, differential effects: a quasi-experimental study of China's vehicle emission policies. *Transportation Research Part D: Transport and Environment*, 158, 105454. <https://doi.org/10.1016/j.trd.2026.105454>
18. Wang, C., Meng, L., Zeng, X.^{*}, **Xie, H.** (2026). Multi-dimensional drivers and their interactions on carbon sink capacity in tropical forests: A case study of Xishuangbanna. *Environment, Development and Sustainability*. In press. <https://doi.org/10.1007/s10668-026-07773-9>
17. Li, H., Chen, B., **Xie, H.**, Yu, H., Dai, M., Xue, M., Long, X., Wang, J., Han, Z., Sun, M., Wang, J., Guo, Y., Lin, J., Wang, Y.^{*}. (2026). Coal phaseout undermines co-benefits of fly ash as a clinker substitute. *the Innovation*, 7, 101496. <https://doi.org/10.1016/j.xinn.2026.101496>
16. Shui, B.^{*}, Cai, Z., **Xie, H.**, Liu, F., Luo, X.^{*}. (2026). Harnessing Policy Synergies to Decarbonize China's Light-Duty Passenger Vehicle Fleet. *Journal of Cleaner Production*, 538, 147236. <https://doi.org/10.1016/j.jclepro.2025.147236>
15. Long, X., Chen, B., Dai, M., **Xie, H.**, Chen, P., Meng, J., Wang, Y.^{*}. (2025). Ambitious climate targets can make the phaseout of India's coal-fired power plants cost-beneficial. *Nature Communications*, 16, 11594. <https://doi.org/10.1038/s41467-025-66580-4>
14. **Xie, H.**[#], Li, Y.[#], Wang, C.^{*}. (2025). Decarbonizing inland waterway transport: scenario analysis and mitigation strategy for Jiangxi Province, 2019–2050. *ENGINEERING Environment*, 19(8), 113. <https://doi.org/10.1007/s11783-025-2033-4>
13. Zhang, Y., Dai, M.^{*}, Chen, B., **Xie, H.**, Li, Z., Wang, Y.^{*}. (2025). A Product-based Life-cycle Integration Platform for sustainable plastic bottles from agricultural biowaste. *Chemical Engineering Journal*, 524, 169320. <https://doi.org/10.1016/j.cej.2025.169320>
12. Bai, Y., Li, H., Chen, B., **Xie, H.**, Wang, Y.^{*}. (2025). Managing nitrogen metabolism of animal husbandry and aquaculture could mitigate nitrogen threat in the Main Cities of Yellow River Delta. *Cleaner Environmental Systems*, 17, 100273. <https://doi.org/10.1016/j.cesys.2025.100273>
11. Wang, C., Kong, Y., Lu, X., **Xie, H.**, Teng, Y., Zhan, J.^{*}. (2024). Rethinking Regional High-Quality Development Pathways from a Carbon Emission Efficiency Perspective. *Land*, 13(9), 1441. <https://doi.org/10.3390/land13091441>
10. Li, H., Bian, Y., Liu, M., Lin, J., Dai, M., **Xie, H.**, Yu, H., Chen, B., Xue, M., Li, Z., Yin, J., Xue, L.^{*}. (2024). CO₂ mineralization and utilization by tailings sand in China for potential carbon sinks and spatial project layout. *Resources, Conservation and Recycling*, 206, 107598. <https://doi.org/10.1016/j.resconrec.2024.107598>
9. Xie, W., Yang, X., Han, Z., Sun, M., Li, Y., **Xie, H.**, Yu, H., Chen, B., Fath, B.^{*}, Wang, Y.^{*}. (2024). Urban sector land use metabolism reveals inequalities across cities and inverse virtual land flows. *Resources, Conservation and Recycling*, 202, 107394. <https://doi.org/10.1016/j.resconrec.2023.107394>

8. Fang, Y., **Xie, H.**, Chen, B., Han, Z., An, D., Cai, W., Zhang, W., Wang, Y.*. (2024). Achieving carbon neutrality in Shanghai's municipal wastewater treatment sector requires coordinated water conservation and technical improvement. *Journal of Cleaner Production*, 443, 141134. <https://doi.org/10.1016/j.jclepro.2024.141134>
7. Han, Z., Xie, W., Yu, H., **Xie, H.**, Li, Y., Wang, Y.*. (2024). Rethinking industrial land-use in American rust cities towards sustainability based on a block-level model. *Journal of Environmental Management*, 352, 120067. <https://doi.org/10.1016/j.jenvman.2024.120067>
6. Dai, M., Sun, M., Chen, B., **Xie, H.**, Zhang, D., Han, Z., Yang, L., Wang, Y.*. (2023). Advancing sustainability in China's pulp and paper industry requires coordinated raw material supply and waste paper management. *Resources, Conservation and Recycling*, 198, 107162. <https://doi.org/10.1016/j.resconrec.2023.107162>
5. Gao, Y., Yi, Y.*, Chen, K., **Xie, H.** (2023). Simulation of suitable habitats for typical vegetation in the Yellow River Estuary based on complex hydrodynamic processes. *Ecological Indicators*, 154, 110623. <https://doi.org/10.1016/j.ecolind.2023.110623>
4. Liang, Z., Deng, H., **Xie, H.**, Chen, B., Sun, M., Wang, Y.*. (2023). Rethinking the paper product carbon footprint accounting standard from a life-cycle perspective. *Journal of Cleaner Production*, 393, 136352. <https://doi.org/10.1016/j.jclepro.2023.136352>
3. **Xie, H.**, Li, Y.*. (2022). An ecological water replenishment model of urban lake riparian plant restoration based on the groundwater – vegetation interactions. *Ecological Engineering*, 176, 106510. <https://doi.org/10.1016/j.ecoleng.2021.106510>
2. **Xie, H.**, Yi, Y.*, Hou, C., Yang, Z. (2020). In situ experiment on groundwater control of the ecological zonation of salt marsh macrophytes in an estuarine area. *Journal of Hydrology*, 585, 124844. <https://doi.org/10.1016/j.jhydrol.2020.124844>
1. Yi, Y.*, **Xie, H.**, Yang, Y., Zhou, Y., Yang, Z. (2020). Suitable habitat mathematical model of common reed (*Phragmites australis*) in shallow lakes with coupling cellular automaton and modified logistic function. *Ecological Modelling*, 419, 108938. <https://doi.org/10.1016/j.ecolmodel.2020.108938>

In Chinese:

5. Wang, C., **Xie, H.**, Zhang, X., Zeng, X.*. (2025). Spatiotemporal variations of carbon source-sink matching in the Beijing-Tianjin-Hebei urban agglomeration. *Chinese Journal of Ecology (生态学杂志)*, 44 (04), 1343-1354. (In Chinese). <https://doi.org/10.13292/j.1000-4890.202504.041>
4. Wang, S., Chen, B., Dai, M., **Xie, H.**, Zhang, D., Sun, M., Wang, Y.*. (2025). Carbon footprint accounting and impact factor analysis of China's bioeconomy. *Environmental Engineering (环境工程)*, 43(8), 233-243. <https://doi.org/10.13205/j.hjgc.202508022>
3. **Xie, H.**, Liu, J., Gao, Y., Li, Q., Gao, J., Yang, S., Zhu, G.*. (2022). Method and Application of Regional Highway Resources and Environment Carrying Capacity Evaluation. *Highway (公路)*, 67 (09), 407-412.
2. Yi, Y.*, **Xie, H.**, Song, J., Yang, Z. (2021). Simulation of salt marsh vegetation community's suitable habitat in Yellow River Estuary I: theory. *Journal of Hydraulic Engineering (水利学报)*, 52(03), 255-264. <https://doi.org/10.13243/j.cnki.slxb.20200542>

1. Yi, Y.^{*}, **Xie, H.**, Song, J., Yang, Z. (2021). Simulation of salt marsh vegetation community's suitable habitat in Yellow River Estuary II :application. *Journal of Hydraulic Engineering (水利学报)*, 52 (04), 401-408. <https://doi.org/10.13243/j.cnki.slxb.20200544>

In revision and under review:

1. Xie, G., Bai, Y., Wang, Q., **Xie, H.**^{*}. Rethinking Urbanization under the Socio - Economic - Ecosystem Nexus: Insights from an Emergy Perspective in Shandong Province, China. (Under review for *Ecological Modelling*)
2. Fu, Y., Li, H., Chen, B., **Xie, H.**, Dai, M., Han, Z., Sun, L., Xue, M., Wang, Y.^{*}. Spatial mismatch of biomass supply constrains biochar-based decarbonization in China's iron and steel industry. (Major revision for *Resources, Conservation & Recycling*)
3. Sun, C., Han, Z., **Xie, H.**, Li, H., Chen, B., Zhao, Y., Wang, Y.^{*}. Spatial Mismatch Embedded in the UK's Sectoral Land Use Efficiency. (Submitted to *Resources, Conservation & Recycling*)
4. Han, Z., Chen, B., Chen, Z., Shi, Z., Wang, S., Song, Y., **Xie, H.**, Li, H., Sun, C., Meng, J., Wu, L.^{*}, Wang, Y.^{*}. Location-sector pairing redefines sectoral flood exposure and network propagation across Chinese cities. (Submitted to *Nature Communications*)
5. Wang, Z., Li, Y.^{*}, **Xie, H.**, Zou, T., Yang, Y. Spatial heterogeneity of life cycle carbon emissions from hydrogen production in China. (Submitted to *Energy*)
6. Li, H., Fu, Y., Dai, M., **Xie, H.**, Han, Z., Xue, M., Chen B.^{*}, Song, X., Sun, C., Wang, S., Cao, D., Wang, Y.^{*}. Supply-limited substitutes undermine cement decarbonization in China. In preparation.
7. Song, Y.[#], Han, Z.[#], **Xie, H.**[#], Chen, B., Wang, Y.^{*}. Local final demand drives China's sectoral land-use-induced biodiversity loss. In preparation.

Reports, books and standards:

1. United Nations Environment Programme (2024). Global Bioeconomy Assessment: Coordinated Efforts of Policy, Innovation, and Sustainability for a Greener Future. <https://wedocs.unep.org/20.500.11822/45332>
2. China Environment Publishing Group (2020). Theory and Practice of Risk Analysis and Emergency Capacity Planning for Oil Spills on Water. ISBN: 978-7-0000-0000-0
3. Chinese Group Standard of China Institute of Navigation (2023). Guidelines for the construction planning of ship oil spill response capacity of inland water (T/CIN 025-2023). https://www.cinnet.cn/zh-hans/system/files/files/25.nei_he_chuan_bo_yi_you_ying_ji_neng_li_jian_she_gui_hua_bian_zhi_zhi_nan_.pdf

AWARDS

Outstanding Graduate, Fudan University (2026)

Joint PhD Training Program for State-constructed High-level Universities, China Scholarship Council (2024)

Second-class Academic Scholarship, Fudan University (2024, 2025)

Third Prize, Excellent Consulting Achievement Award in Port and Waterway Engineering, China Waterborne Transport Construction Association (2023)

Outstanding Graduate, Beijing Normal University (2020)

CONFERENCES & PRESENTATIONS

2025 – 12th International Conference on Industrial Ecology, Singapore – Oral Presentation

2024 – Nature Conference on Air Pollution and Climate Change, Beijing – Poster Presentation

2023 – International Conference on Resource Sustainability, United Kingdom – Oral Presentation

2018 – 1st International Student Forum on Environmental Ecology, Beijing – Poster Presentation

TECHNICAL SKILLS

Analysis: Techno-economic assessment; Life Cycle Assessment; Prospective LCA (premise); Material Flow Analysis; Emission inventory; Spatial Analysis (ArcGIS)

Modeling: Integrated Assessment Model (GCAM); Multi-Agent modeling (AnyLogic); Hydrodynamic modeling (MIKE 21, Visual MODFLOW)

Coding: Matlab, R, Python

SERVICE

Journal referee: *Nature Cities; Environmental Science & Technology; Resources, Conservation and Recycling; Applied Energy; Journal of Cleaner Production; Ecological Economics; Journal of Environmental Management; Scientific Data; Geography and Sustainability; Ecological Indicators; iScience; Journal of Hydrology: Regional Studies; Regional Studies in Marine Science; Journal of Sea Research; Heliyon; Energy, Ecology and Environment; Results in Engineering; Bioresource Technology Reports; Environmental Development; Research in Transportation Business & Management; Unconventional Resources; Energy Nexus; Environmental Nexus; Atmospheric Environment: X*

REFEREES

Prof. Yanyan Cheng, Nanyang Technological University, yanyan.cheng@ntu.edu.sg

Prof. Yutao Wang, Fudan University, yutaowang@fudan.edu.cn

Prof. Yang Guo, Beijing Normal University, guoyang@bnu.edu.cn